

Visualising High-dimensional Data with R

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Session 2

Outline

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time	topic
30	Understanding clusters in data using visualisation
30	Building better classification models with visual input

Clustering

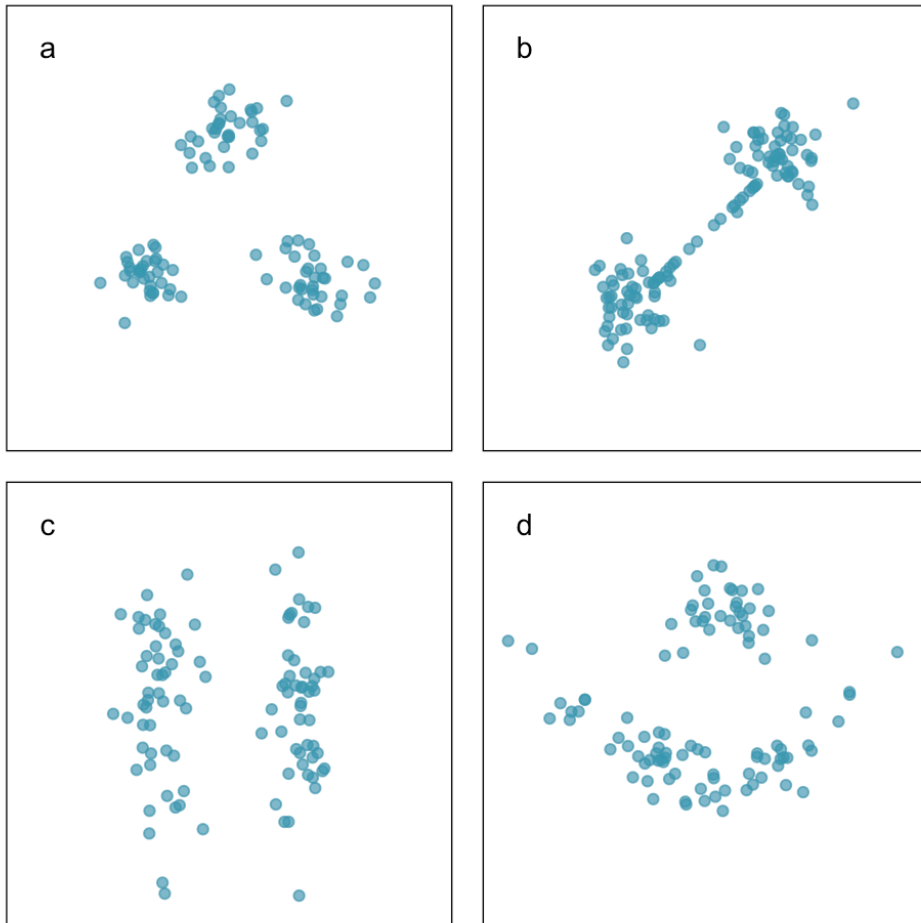
Method 1: Spin-and-brush

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DEMO

What are clusters?

Ideal thinking of neatly separated clusters, but it is rarely encountered in data

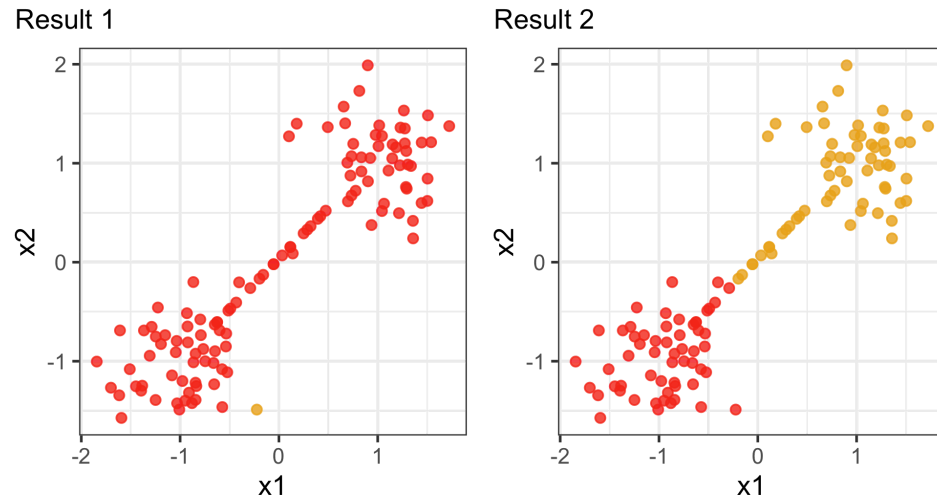


Objective is to organize the cases into groups that are similar in some way. You need a measure of similarity (or distance).

Why visualise? Which is the better?

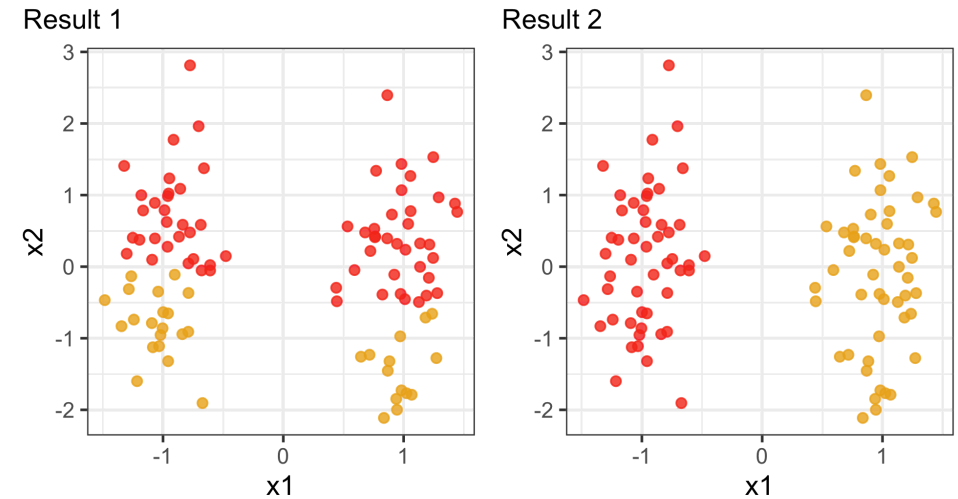
Nuisance cases

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Nuisance variables

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To decide on a best result, you need to see how it **divides the data into clusters**. The cluster statistics, like dendrogram, or cluster summaries, or gap statistics might all look good but the result is bad. You need to see the **model in the data space**!

Model-based clustering (1/3)

Model-based clustering fits a multivariate normal mixture model to the data.

$$\Sigma_k = \lambda_k D_k A_k D_k^\top, \quad k = 1, \dots, g$$

where

Σ_k is the variance-covariance of cluster k ,

g = number of clusters,

D_k describes the orientation of a cluster,

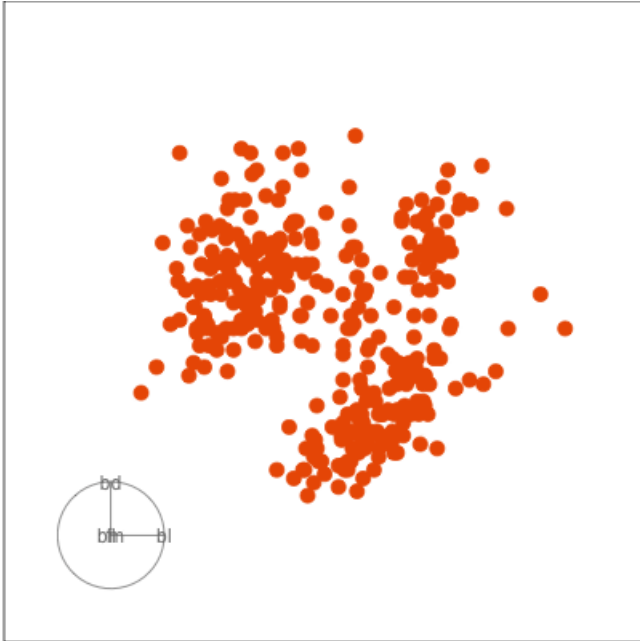
A_k describes the variance in different variables,

λ_k is an overall size.

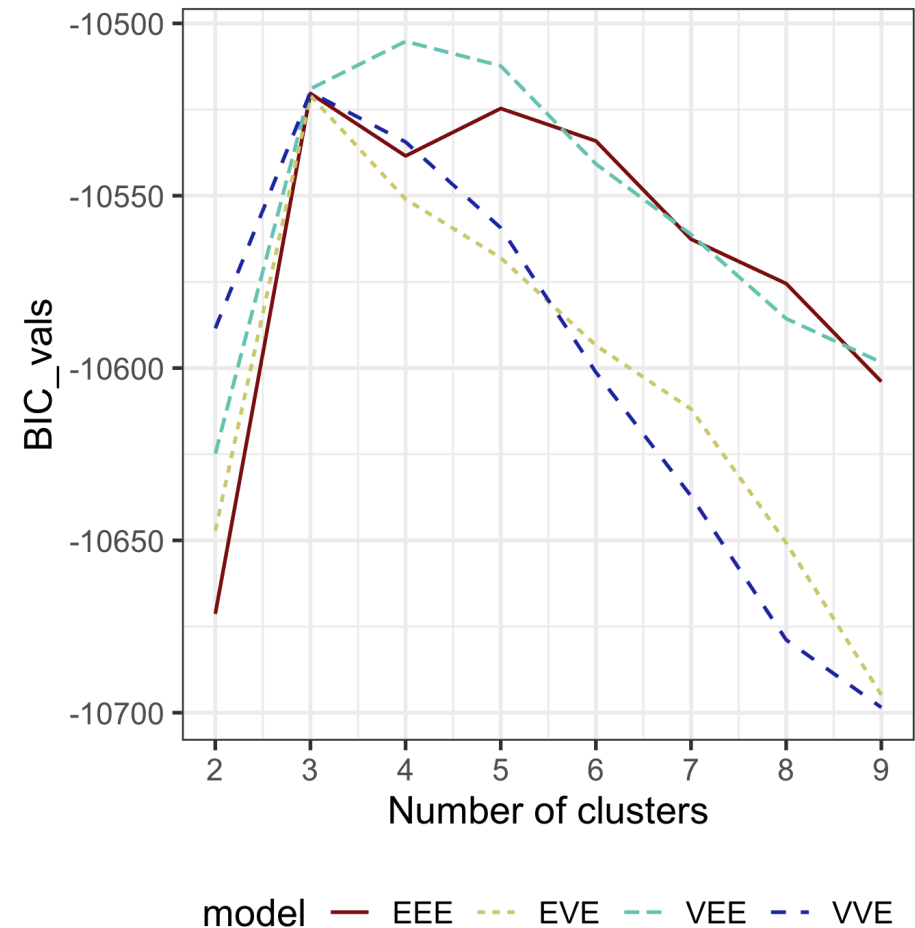
Model-based clustering (2/3)

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Clustering this data. What do you expect?



Can we assume the shape of the clusters is elliptical?

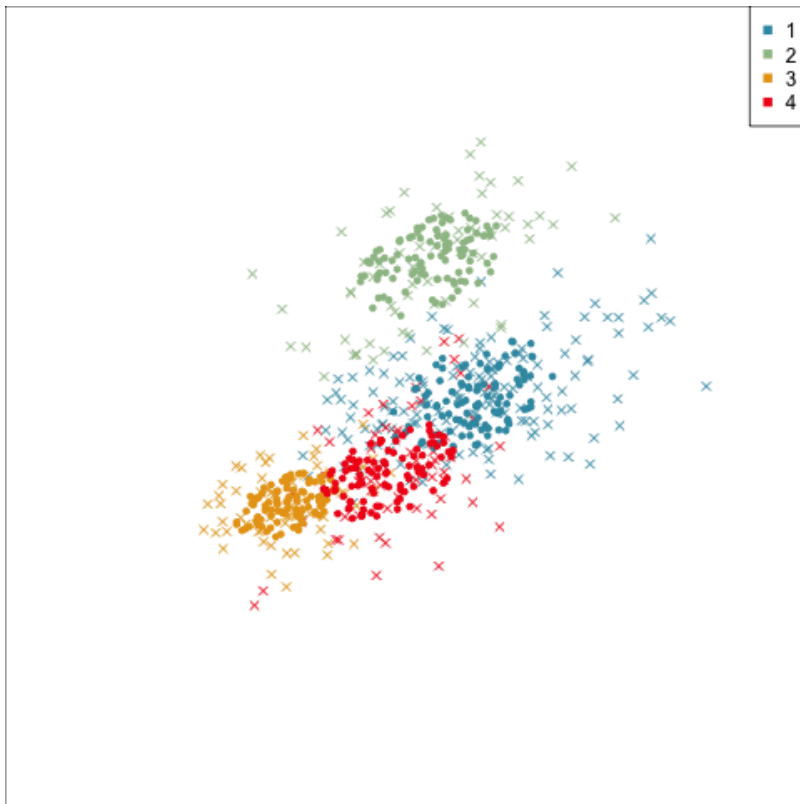


Model-based clustering (3/3)

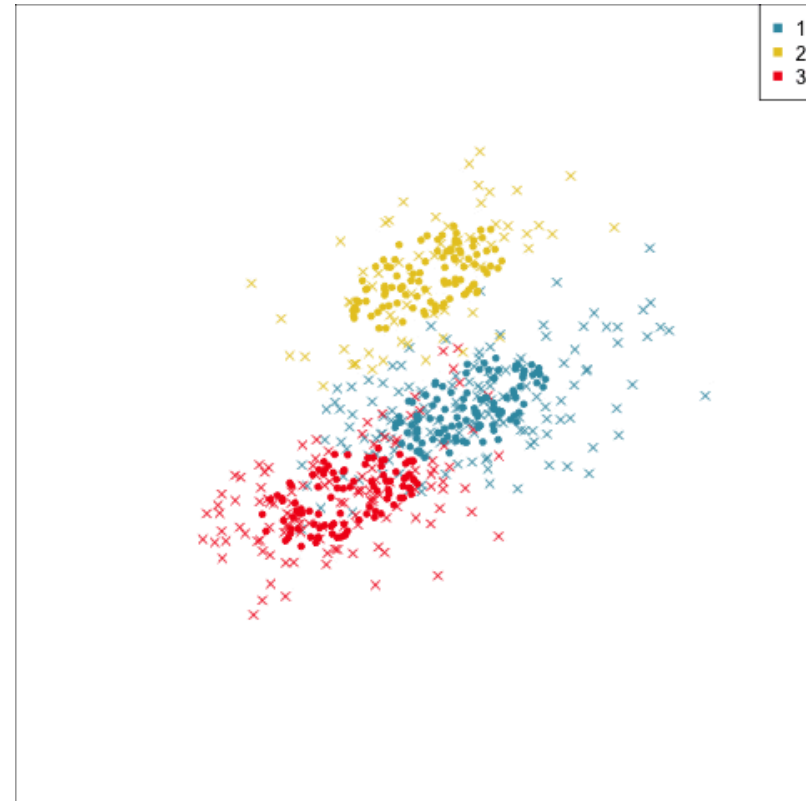
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Four-cluster VEE



Three-cluster EEE



How do you draw ellipses in high-d?

Extract the estimated model parameters

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```
      [,1] [,2] [,3]  
bl    39   48   49  
bd    18   15   18  
fl   190  217  196  
bm 3693 5076 3754
```

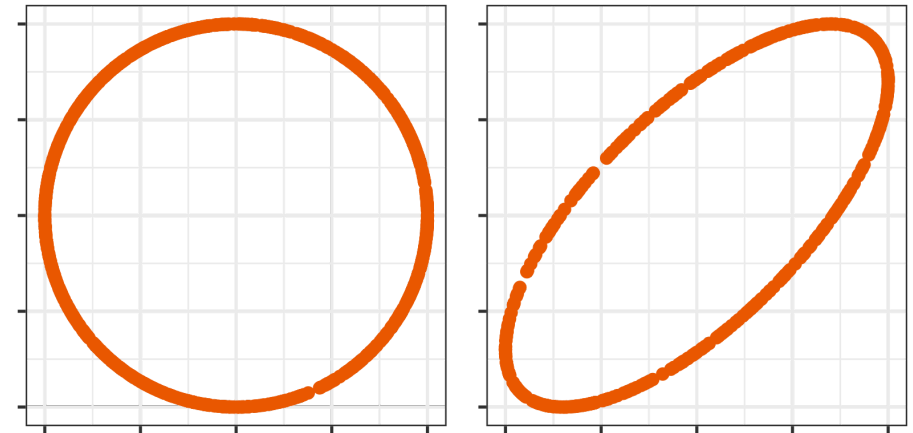
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```
      bl    bd    fl    bm  
bl    8.4    1.6    8.3    755  
bd    1.6    1.2    3.5    318  
fl    8.3    3.5   42.5   1751  
bm 754.7 318.4 1751.2 211467
```

Generate data that represents the ellipse(s) to overlay on the data.

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- Sample points uniformly on a pD sphere
- Transform into an ellipse using the inverse variance-covariance matrix



Your turn

Use the spin-and-brush approach to extract the clusters from the **c1** data.

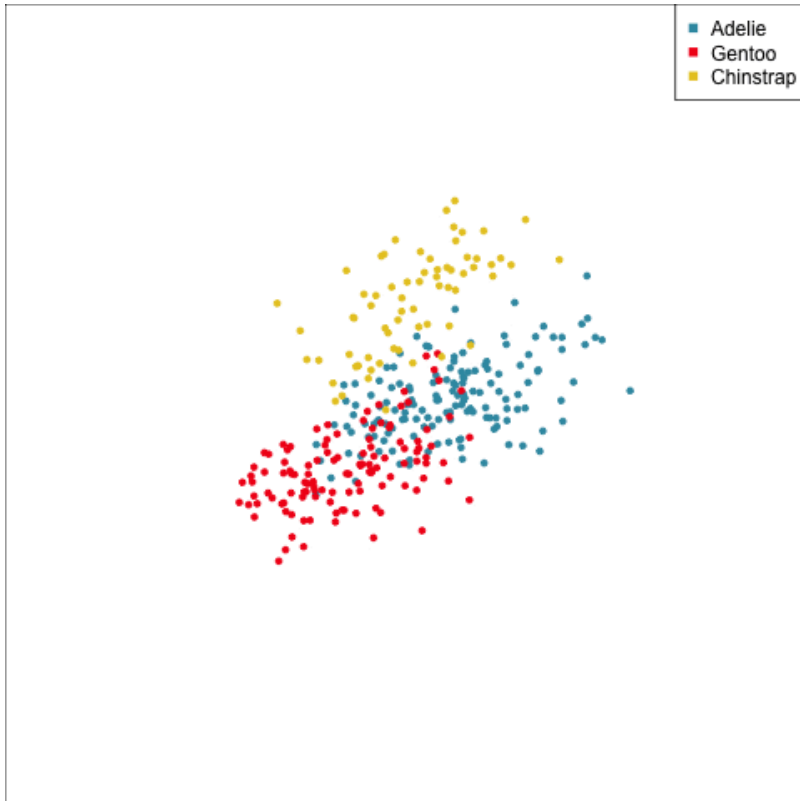
Classification

What should you visualise?

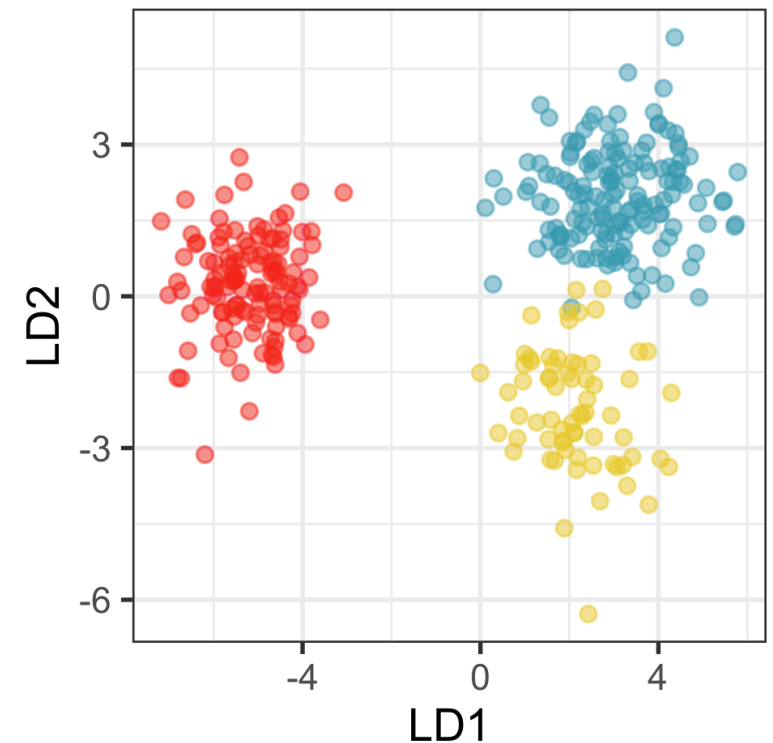
- Understand any clustering related to the known classes.
- Obtain a sense of where **boundaries** might be placed.
- Examine where the fitted model fits the data well, and where poorly.
- Understand the **misclassifications**, whether they are reasonable given uncertainty in the data, or due to an ill-fitting or poorly specified model.
- Understand what can happen with model fitting and pattern recognition with sparse data.

Example: Linear DA

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Linear discriminant analysis is the ideal classifier for this data.



● Adelie ● Chinstrap ● Gentoo

Random forests (1/2)

A random forest is the simplest classifier to fit for complicated boundaries. It is built from multiple trees generated by randomly sampling the cases and the variables. The random sampling (with replacement) of cases has the fortunate effect of creating a training (“in-bag”) and a test (“**out-of-bag**”) sample for each tree computed. The most beautiful results are obtaining diagnostics that help us to assess the model are the **votes**, the measure of **variable importance**, and the **proximity matrix**.

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Call:

```
randomForest(formula = cause ~ ., data =  
bushfires_sub, importance = TRUE)  
Type of random forest: classification  
Number of trees: 500  
No. of variables tried at each split: 7
```

OOB estimate of error rate: 11%

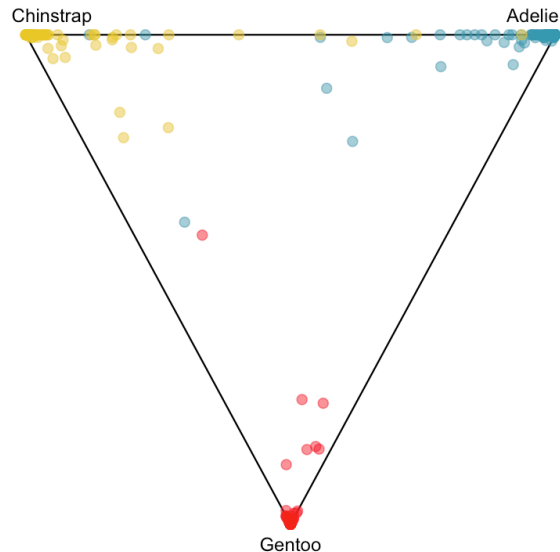
Confusion matrix:

	accident	arson	burning_off	lightning
accident	73	3	0	62
arson	11	8	1	17
burning_off	3	0	3	3
lightning	14	0	0	823

	class.error
accident	0.471
arson	0.784
burning_off	0.667
lightning	0.017

Random forests (2/2)

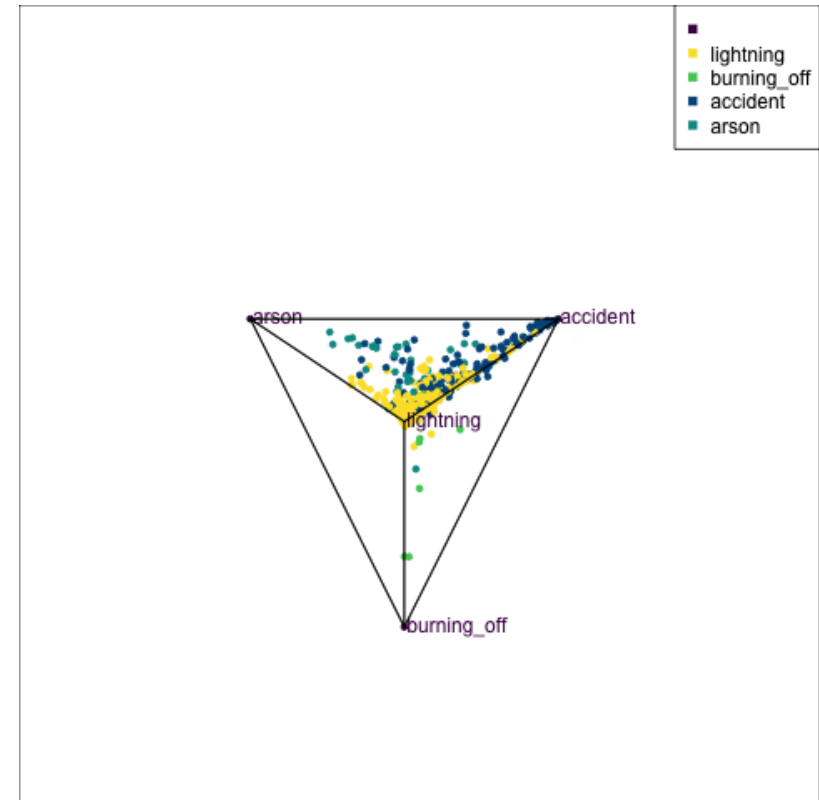
The **votes matrix** can be considered to be predictive probabilities, where the values for each observation sum to 1. With 3 classes it is a 2D triangle. For 4 or more classes it is a **simplex** and can be examined in a tour.



Votes matrix for the random forest fit on penguins

Votes matrix for bushfire model fit

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Exploring misclassifications

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DEMO

Your turn

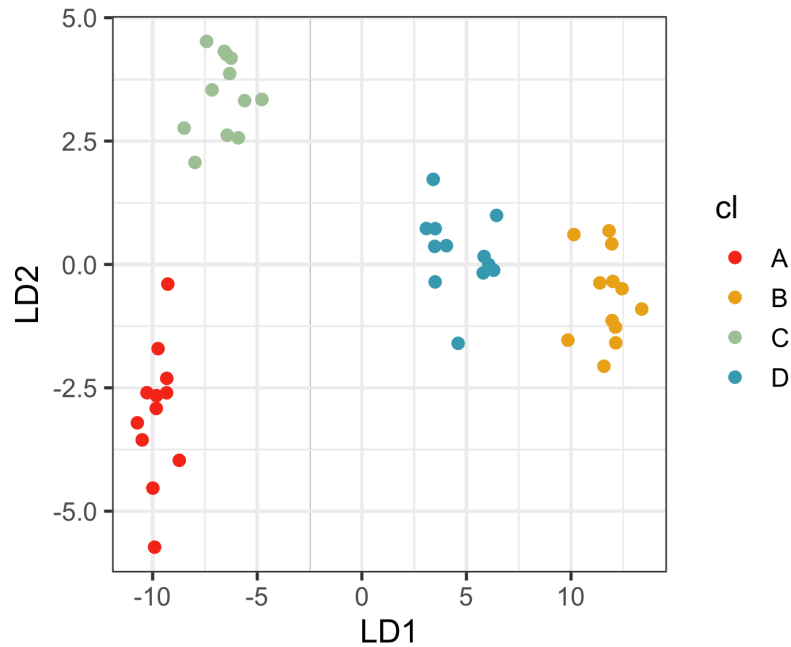
Explore the misclassifications in the **random forest fit** of the **penguins** data, using the code provided in the **slides2.R** file.

Cautions about high-dimensions

Space is big.

What might appear to be **structure** is only **sampling variability**.

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$$n = 48, p = 40$$

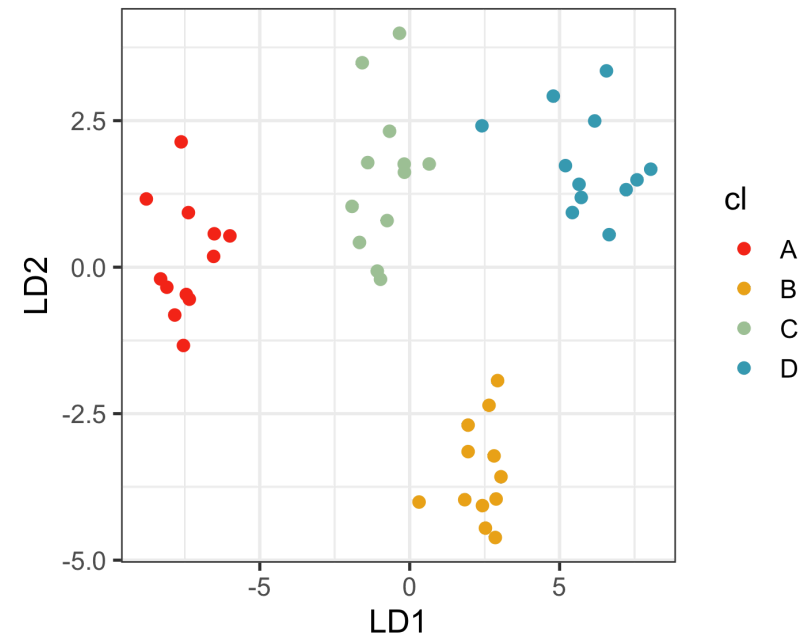
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Permutation is your friend, for high-dimensional data analysis.

Permute the class labels.

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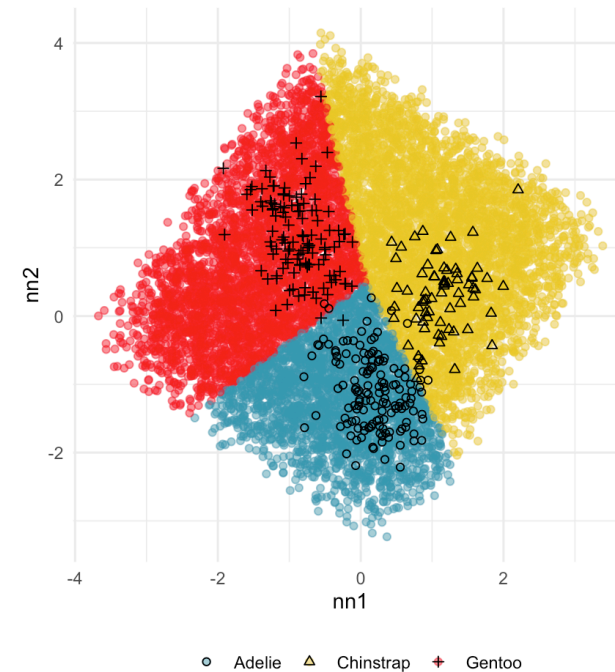
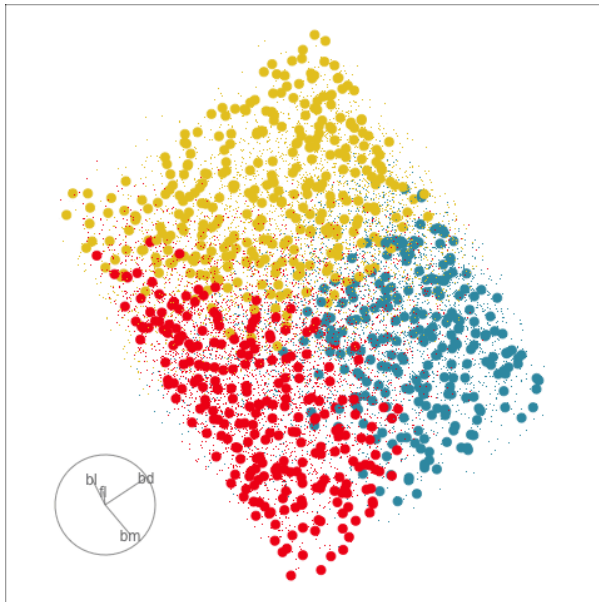
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Other compelling pursuits

Explore and compare the boundaries of different models using the slice tour.

Dissect and explore the operation of a neural network.

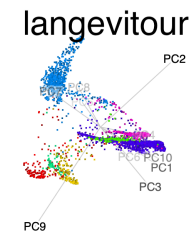
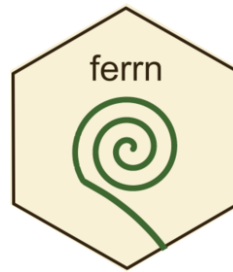


Where to learn more

All of the material presented today comes from

Cook and Laa (2026) Interactively exploring high-dimensional data and models in R

Software:



End of session 2



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